

1 The image shows a section of a recombinant DNA molecule.



(a) (i) Which enzyme is used in the production of recombinant DNA?

(1)

- ☐ **A** acetylcholinesterase
- ☐ **B** amylase
- ☐ **C** restriction endonuclease
- ☐ **D** RNA polymerase

(ii) DNA is present in bacterial cells.

Where is DNA found in a bacterial cell?

(1)

- ☐ **A** nucleus only
- ☐ **B** plasmid only
- ☐ **C** plasmid and cytoplasm
- ☐ **D** plasmid and nucleus

(iii) Which bond joins nucleotides together in a **single** strand of DNA?

(1)

- ☐ **A** hydrogen
- ☐ **B** ionic
- ☐ **C** peptide
- ☐ **D** phosphodiester

(b) Describe how genes can be switched **off** by transcription factors.

(3)

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(Total for Question 1 = 6 marks)